MATH102BKK · HARBOUR.SPACE BANGKOK

Functions

From "formula" to "function": a rule is not enough — you must say where it starts and where it lands.

Session	Date	Today
3 of 15	Wed 9 Sep 2026	Domain, codomain, undoing

COURSE OUTLINE

Where session 3 sits

1 · Course overview and school-maths refresher	2 · Sets and set operations
3 · Functions	4 · Function families
5 · Theorem proving	6 · Mathematical induction
7 · Elementary combinatorics and binomial theorem	8 · Midterm
9 · GCD and divisibility	10 · Modular arithmetic
11 · Vectors	12 · Polynomials
13 · Complex numbers	14 · Course recap and exam preparation
15 · Final exam	

TODAY Domain and codomain, injective/surjective/bijective, composition and inverses, and the unit circle.

Set theory, in one page

OBJECTS	
set	Unordered, no repeats: $\{1,2,2\} = \{1,2\}$
$x \in A$	Element of a set
$\{1,2,3\}$	Roster; or set-builder $\{x \in \mathbb{Z} : x^2 < 10\}$
$\varnothing$	Empty: no elements. $\{\varnothing\}$ has one
$A \subseteq B$	Subset: all of A lies in B
$A = B$	$A \subseteq B$ and $B \subseteq A$: double inclusion
$\mathcal{P}(A)$	Power set: all subsets. $\lvert\mathcal{P}(A)\rvert = 2^{\lvert A\rvert}$

OPERATIONS	
$A \cup B$	Union: A or B (inclusive)
$A \cap B$	Intersection: in both. Empty: disjoint
$A \setminus B$	Difference: A without B. Asymmetric
A^c	Complement $U \setminus A$; name U first
$A \triangle B$	Symmetric difference: exactly one
De Morgan	$(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$
$A \times B$	Ordered pairs (a,b); $\lvert A \times B\rvert = \lvert A\rvert\lvert B\rvert$

TODAY'S QUESTION Is x^2 a function? Can you run it backwards?

CONCEPT I

A function is three things, not one

DEFINITION A function $f : A \to B$ is three things at once:

- a **domain** A
- a **codomain** B
- a rule giving **each** $a \in A$ **exactly one** $f(a) \in B$

Its **graph** is a subset of $A \times B$, with each a occurring in exactly one pair.

IS $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$. Every real has exactly one square, and it is real.

IS NOT

$g : \mathbb{R} \to \mathbb{R}$, $g(x) = 1/x$: nothing at $x = 0$. *Not total.*

The relation $y^2 = x$ on $\mathbb{R}$: two possible y values when $x = 4$, and none when $x = -1$. *Not a function.*

CONCEPT I

The codomain is declared; the image is computed

DEFINITION

$$f(A) = \{\, f(a) : a \in A \,\} \subseteq B$$

$$f^{-1}(S) = \{\, a \in A : f(a) \in S \,\}$$

The first is the **image**, the second the **preimage** of $S \subseteq B$.

WORKED

$f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$.

Codomain $\mathbb{R}$, image $[0, \infty)$. So -1 sits in the codomain but never in the image.

$f^{-1}(\{4\}) = \{-2, 2\}$, while $f^{-1}([-1, 0)) = \varnothing$.

NOTATION WARNING $f^{-1}(S)$ takes a **set** and returns a **set**. It exists for every function. It is not the inverse function f^{-1} we meet after the break, which needs a bijection.

Which of the three ingredients is missing?

ACTIVITY 1 GROUPS OF THREE 12 MIN

For each line: is it a function? If not, name **exactly** which requirement fails, and give the input that breaks it.

1. $f : \mathbb{Z} \to \mathbb{Z}$, $f(n) = n/2$
2. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \sqrt{x}$
3. $f : \mathbb{Q} \to \mathbb{Z}$, $f(p/q) = p$
4. $f : \mathbb{R} \to [0, \infty)$, $f(x) = x^2$
5. $f : \mathbb{R} \to \mathbb{R}$ defined by the condition $f(x)^2 = x$

Share: each group takes one line and names the failure — *not total*, *not single-valued*, *lands outside the codomain*, or *fine*.

Name the requirement that fails

RULE	VERDICT	THE BREAKING INPUT
1 · $\mathbb{Z} \to \mathbb{Z}$, $n/2$	Leaves the codomain	$f(3) = 1.5 \notin \mathbb{Z}$.
2 · $\mathbb{R} \to \mathbb{R}$, $\sqrt{x}$	Not total	No value at $x = -1$.
3 · $\mathbb{Q} \to \mathbb{Z}$, $p/q \mapsto p$	Not well defined	$\frac{1}{2} = \frac{2}{4}$, yet the rule returns 1 and 2.
4 · $\mathbb{R} \to [0, \infty)$, x^2	A function	Total, single-valued, lands in $[0, \infty)$.
5 · $\mathbb{R} \to \mathbb{R}$, $f(x)^2 = x$	No such function	At $x = -1$ we would need $f(-1)^2 = -1$.

LINE 3 IS THE NEW IDEA The input is a *rational number*, not a pair of integers. A rule using p and q must agree on every fraction naming that number.

Injective, surjective, bijective — three questions about hitting

DEFINITION

Injective: $f(a_1) = f(a_2) \implies a_1 = a_2$. Nothing in B is hit twice.

Surjective: $\forall b \in B\ \exists a \in A$ with $f(a) = b$. Nothing in B is missed.

Bijective: both. Every b hit exactly once.

SAME RULE, OPPOSITE ANSWERS

$f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$: injective fails at $f(-2) = f(2)$; surjective fails at -1.

$f : [0, \infty) \to [0, \infty)$, $f(x) = x^2$: **both** hold. Nothing about the rule changed.

Each of the four claims has one fixed opening move

THE FOUR TEMPLATES

Injective: assume $f(a_1) = f(a_2)$, derive $a_1 = a_2$.

Not injective: exhibit one pair $a_1 \neq a_2$ with equal images.

Surjective: take arbitrary $b \in B$, *construct* an a and check $f(a) = b$.

Not surjective: exhibit one b and *prove* nothing maps to it. ∎

WORKED: $f : \mathbb{R} \to \mathbb{R},\ f(x) = 3x - 5$

Injective: $3a_1 - 5 = 3a_2 - 5 \Rightarrow 3a_1 = 3a_2 \Rightarrow a_1 = a_2$.

Surjective: given y, set $x = \frac{y+5}{3}$; then $f(x) = 3 \cdot \frac{y+5}{3} - 5 = y$. So f is a bijection.

ACTIVITY 2

Classify, then prove it

ACTIVITY 2 GROUPS OF THREE 15 MIN

For each: classify as injective, surjective, both or neither — then prove it with a template.

1. $f : \mathbb{Z} \to \mathbb{Z}$, $f(n) = 2n + 1$
2. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3$
3. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$
4. $f : \mathbb{N} \to \mathbb{N}$, $f(n) = n + 1$
5. $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $f(a, b) = a - b$
6. $f : \mathbb{R} \to (0, \infty)$, $f(x) = 2^x$

Share: one group gives a *positive* proof, one a counterexample. The room votes first.

Six classifications, each with its reason

f	INJECTIVE?	SURJECTIVE?	THE REASON
1 · $\mathbb{Z} \to \mathbb{Z}$, $2n+1$	Yes	No	Strictly increasing; but $2n+1$ is odd, so 2 is missed.
2 · $\mathbb{R} \to \mathbb{R}$, x^3	Yes	Yes	Bijective, inverse $\sqrt[3]{y}$.
3 · $\mathbb{R} \to \mathbb{R}$, x^2	No	No	$f(-1) = f(1)$; and -1 is never attained.
4 · $\mathbb{N} \to \mathbb{N}$, $n+1$	Yes	No	Nothing maps to 1 (taking $\mathbb{N}$ to start at 1).
5 · $\mathbb{Z}^2 \to \mathbb{Z}$, $a-b$	No	Yes	$f(1,1) = f(2,2)$; every k comes from $(k,0)$.
6 · $\mathbb{R} \to (0,\infty)$, 2^x	Yes	Yes	Bijective, inverse $\log_2 y$.

WATCH OUT

"The function x^2" names no function at all

DOMAIN	CODOMAIN	INJECTIVE?	SURJECTIVE?	SO IT IS...
$\mathbb{R}$	$\mathbb{R}$	No (± 2)	No (-1)	neither
$[0, \infty)$	$\mathbb{R}$	Yes	No (-1)	injective only
$\mathbb{R}$	$[0, \infty)$	No (± 2)	Yes	surjective only
$[0, \infty)$	$[0, \infty)$	Yes	Yes	bijective

TRAP One rule, four functions, four different answers. A question asking "is x^2 injective?" is not yet a question.

FIX Write $f : A \to B$ **before** you write the rule. Every answer you write this term starts that way.

PART II

Composing, and undoing

Functions are things you can chain. Some chains can be reversed — and the condition for that is exactly the vocabulary from the first hour.

2

CONCEPT II

Composition needs the codomain of one to be the domain of the next

DEFINITION The **composition** of $f : A \to B$ with $g : B \to C$ is

$$(g \circ f)(a) = g\big(f(a)\big), \qquad g \circ f : A \to C$$

Read it right to left: f runs first. If the middle sets do not match, the composition does not exist.

ORDER IS NOT A DETAIL

$f(x) = x + 1$, $g(x) = x^2$, both $\mathbb{R} \to \mathbb{R}$.

$$(g \circ f)(x) = (x+1)^2 \qquad (f \circ g)(x) = x^2 + 1$$

At $x = 1$: 4 against 2. So $g \circ f \neq f \circ g$.

CONCEPT II

Injectivity of a chain is inherited by its first link

THEOREM Let $f : A \to B$ and $g : B \to C$. If $g \circ f$ is injective, then f is injective.

PROOF Suppose $f(a_1) = f(a_2)$. Applying g to both sides gives $g(f(a_1)) = g(f(a_2))$, that is $(g \circ f)(a_1) = (g \circ f)(a_2)$. Since $g \circ f$ is injective, $a_1 = a_2$. Hence f is injective. ∎

AND g? Nothing follows. With $A = \{1\}$, $B = \{1, 2\}$, $C = \{1\}$, $f(1) = 1$, $g(1) = g(2) = 1$: the chain is injective, g is not.

A function can be undone exactly when it is a bijection

THEOREM $f : A \to B$ has a $g : B \to A$ with $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$ if and only if f is a bijection. This g is unique: it is f^{-1}.

PROOF ($\Longleftarrow$) Let $b \in B$. Surjectivity gives an a with $f(a) = b$; injectivity says it is the only one. So $g(b) = a$ is well defined, and both composites are identities. ■

TRAP f^{-1} is not $\frac{1}{f}$. For $f(x) = 3x - 5$: $f^{-1}(1) = 2$, while $\frac{1}{f(1)} = -\frac{1}{2}$.

FIX f^{-1} undoes; $\frac{1}{f}$ divides. Same superscript, unrelated operations.

ACTIVITY 3

Find the broken step

ACTIVITY 3 GROUPS OF THREE 14 MIN

Claim. If $g \circ f$ is injective, then g is injective. **"Proof."**

(i) Suppose $g(b_1) = g(b_2)$.
(ii) Write $b_1 = f(a_1)$ and $b_2 = f(a_2)$.
(iii) Then $(g \circ f)(a_1) = (g \circ f)(a_2)$, so $a_1 = a_2$.
(iv) Hence $b_1 = f(a_1) = f(a_2) = b_2$. ■

1. Which of (i)–(iv) is the first unjustified line?
2. Give a counterexample showing the claim is false.
3. Add the smallest hypothesis on f that repairs the proof.
4. State and prove the companion result for $g \circ f$ *surjective*.

Share: one group presents the counterexample, another the repaired statement. The room decides if step 3 is the smallest fix.

Step (ii) assumed what was never given

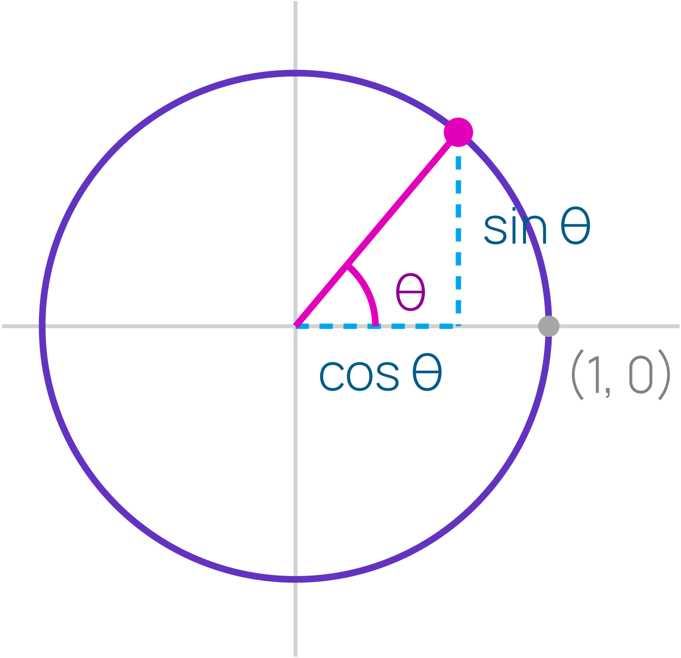
1 · THE FIRST UNJUSTIFIED LINE IS (II) "Write $b_1 = f(a_1)$" assumes every element of B is an output of f — that is surjectivity, never assumed.

2 · COUNTEREXAMPLE $f : \{1\} \to \{1,2\}$, $f(1) = 1$; and $g : \{1,2\} \to \{1\}$, $g(1) = g(2) = 1$. Then $g \circ f$ is injective but g is not.

3 · THE REPAIR Assume f is surjective. Step (ii) is then legitimate and the rest stands.

4 · THE COMPANION RESULT If $g \circ f$ is surjective, so is g: given $c \in C$, pick a with $g(f(a)) = c$; then $f(a)$ is a preimage. $\blacksquare$

Sine and cosine are coordinates, and the identity is Pythagoras



DEFINITION From $(1, 0)$, walk θ counter-clockwise along the unit circle $x^2 + y^2 = 1$. The point you reach is $(\cos\theta,\ \sin\theta)$, and $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$ where $\cos\theta \neq 0$. That distance is θ radians.

WHY THE IDENTITY IS FREE The point lies *on* the circle, so $x^2 + y^2 = 1$. Substitute:

$$\cos^2\theta + \sin^2\theta = 1$$

Not a new fact — the circle's equation, renamed.

Periodicity gives repeated outputs

f	DOMAIN	IMAGE	PERIOD	INJECTIVE THERE?
$\sin$	$\mathbb{R}$	$[-1,1]$	2π	No $(\sin 0 = \sin \pi)$
$\cos$	$\mathbb{R}$	$[-1,1]$	2π	No $(\cos(-\theta) = \cos \theta)$
$\tan$	$\mathbb{R} \setminus \{\frac{\pi}{2} + k\pi\}$	$\mathbb{R}$	π	No (period π)

DEFINITION $p > 0$ is a period of f if $f(\theta + p) = f(\theta)$ for every θ. When there is a smallest positive one, it is the fundamental period.

THE RESTRICTION TRICK, AGAIN $\sin$ cut down to $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \to [-1,1]$ is a bijection, so by the theorem it has an inverse. That inverse is $\arcsin$.

WATCH OUT

The unit circle definition already fixed your units

TRAP

$\sin(30) \neq \frac{1}{2}$.

θ is an arc length, so $\sin(30) \approx -0.988$: the point after walking 30 units round a circle of circumference 2π. What you meant was $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$.

FIX

Convert once, at the start: $\theta_{\text{rad}} = \dfrac{\pi}{180}\,\theta_{\text{deg}}$.

Radians are the default here, in every calculus course, and in every programming language's standard library. Degrees are the special case that needs announcing.

THIRTY SECONDS Convert $45°$, $60°$ and $270°$ to radians. Then: what is $\cos(180)$, and what did whoever wrote it mean?

ACTIVITY 4

Derive the circle without us

ACTIVITY 4 PAIRS 12 MIN

Draw one unit circle. Close every other source — no tables, no phones.

1. Place $\theta = \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}$ and read off $(\cos\theta, \sin\theta)$ from the $30{-}60{-}90$ and $45{-}45{-}90$ triangles.
2. From the picture alone, argue $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$.
3. From the picture alone, argue $\sin(\theta + \pi) = -\sin\theta$.
4. 2π is a period of $\tan$. Is it the *smallest* one? Justify.

Share: one pair draws the circle and defends step 4 at the board.

Everything read off one picture

1 · THE THREE ANGLES

$\frac{\pi}{6} : \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right) \cdot \frac{\pi}{4} : \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right) \cdot \frac{\pi}{3} : \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

2 · EVEN AND ODD Negating θ reflects the point in the x-axis: the first coordinate is unchanged and the second flips. So $\cos(-\theta) = \cos\theta$, $\sin(-\theta) = -\sin\theta$.

3 · ADDING π Adding π sends the point to its antipode, negating both coordinates. Reading off the second gives $\sin(\theta + \pi) = -\sin\theta$.

4 · "A PERIOD" IS NOT "THE PERIOD" No. $\tan(\theta + \pi) = \tan\theta$, so π is smaller — and π is in fact the smallest. Verifying one value never establishes minimality.

EXIT TICKET · 10 MINUTES

On your own, before you leave

1 Choose a domain and codomain that make $x \mapsto x^2 - 4x$ a bijection, and prove both halves.

2 With $f(x) = 2x + 1$ and $g(x) = x^2$ on $\mathbb{R}$: compute $g \circ f$ and $f \circ g$, and give one x where they differ.

3 Derive $\cos \frac{\pi}{3}$ and $\sin \frac{\pi}{3}$ from the unit circle. Recalling them scores zero.

RULE FOR TODAY Every verdict carries its template. "Injective" alone is not an answer; "injective, because $f(a_1) = f(a_2)$ forces $a_1 = a_2$ since..." is.

WRAP-UP

What today was actually about

- → A rule is not a function. **Domain and codomain are part of the object** — change them and every answer changes.
- → Injective and surjective are **claims to be proved**, each with a fixed opening move.
- → A function is invertible exactly when it is a bijection — which is why $\arcsin$ needs a restricted domain.

domain · codomain · image · preimage · bijective · composition · inverse · radian · period

Problem Set 3

Eight injective/surjective classifications, each with a proof or a counterexample, plus five unit-circle values *derived*, not recalled.

Next session: **Function families** — polynomial, exp/log and trig, and how fast each one grows.